

EXPERIMENT - 1

Identification and Observation of Bad Designs

This report documents five examples of bad design observed at VIT University, Vellore. Each entry includes a photograph, description, location, reasons the design is flawed, and a suggested improvement.

Bad Design 1: Exposed Outdoor Electrical Distribution Panel



Description:

An outdoor electrical panel with multiple switches, an isolator, and a Legrand box mounted on a wooden board. Numerous wires hang loose and unsecured from the bottom of the panel, exposed to weather and foot traffic.

Source of Photograph:

Own photograph, captured via GPS Map Camera (GPS-enabled).

Place:

Jimmy Carter Road, VIT University, Vellore, Tamil Nadu 632014, India (Lat 12.970901, Long 79.164399)

Reasons why the design is bad:

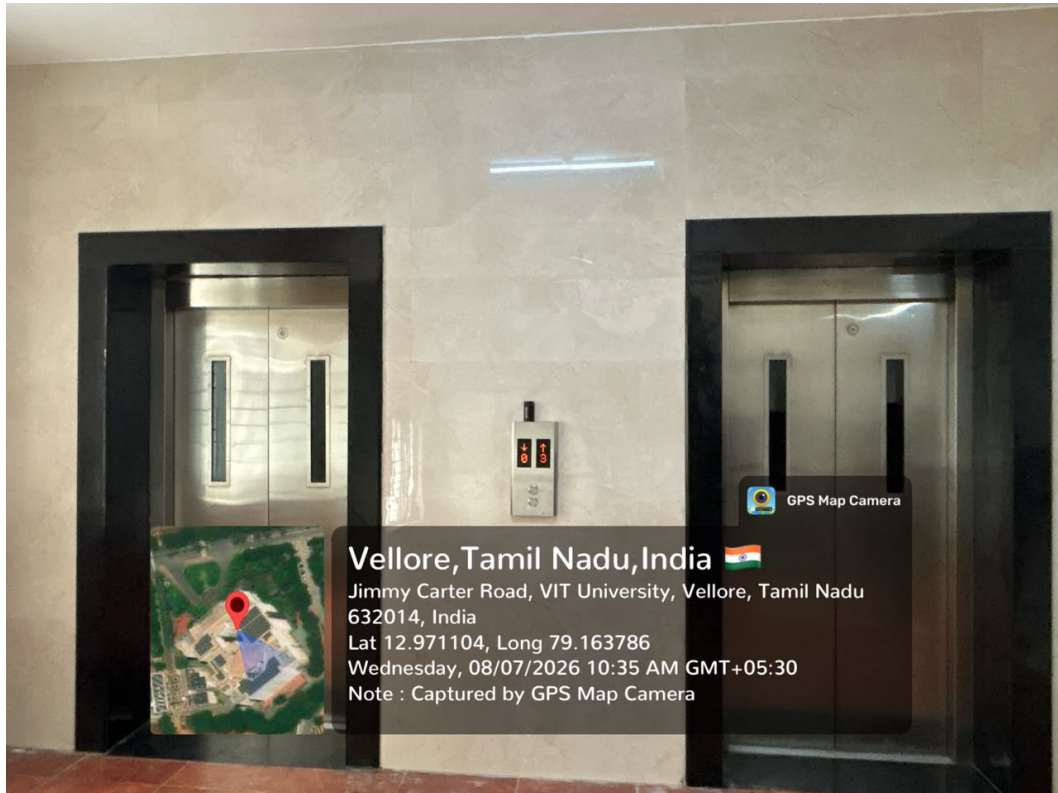
- Exposed, unsecured wiring poses electrocution risk
- No weatherproof enclosure despite outdoor placement

- Loose wires near ground level increase tripping/damage risk
- No visible warning/hazard signage

Suggested Improvement:

- Install a weatherproof (IP-rated) enclosure
- Route and secure wires using conduits or trunking
- Elevate wiring away from ground/foot traffic
- Add hazard warning signage and a restricted-access barrier

Bad Design 2: Elevator Lobby with Shared, Off-Center Indicator Panel



Description:

Two elevators facing each other share a single call/floor indicator panel mounted off to one side, closer to one elevator than the other.

Source of Photograph:

Own photograph, captured via GPS Map Camera (GPS-enabled).

Place:

Jimmy Carter Road, VIT University, Vellore, Tamil Nadu 632014, India (Lat 12.971104, Long 79.163786)

Reasons why the design is bad:

- Panel is not equidistant, making it harder to read from the second elevator
- Only one small display shared between two elevators causes confusion about which car is arriving
- No directional indicator visible from both entrances simultaneously

Suggested Improvement:

- Install separate indicator displays above each elevator door
- Center the shared panel between both elevators if only one is used
- Add audio cues to reduce dependence on visual indicators

Bad Design 3: Unsafe Water Heater Placement in Bathroom



Description:

An electric water heater (geyser) mounted directly above a wash area, with exposed wiring and a switch board on the wall of a wet zone, plus visibly unfinished/peeling wall plaster.

Source of Photograph:

Own photograph, captured via GPS Map Camera (GPS-enabled).

Place:

Jimmy Carter Road, VIT University, Vellore, Tamil Nadu 632014, India (Lat 12.970394, Long 79.164326)

Reasons why the design is bad:

- Electrical wiring and switches exposed in a high-moisture area is a shock hazard
- Water heater positioned too high for safe/easy access to controls
- Poor wall finishing near electrical fittings increases risk of water seepage into wiring

Suggested Improvement:

- Use moisture-proof, sealed wiring and switch covers rated for wet areas
- Relocate switches outside the wet zone
- Properly seal/finish walls around electrical fittings
- Position heater at an accessible, code-compliant height

Bad Design 4: External Fire-Escape Staircase Without Visible Safety Features



Description:

A tall, narrow metal fire-escape staircase attached to the exterior of a multi-story hostel/academic building, with no visible signage or lighting.

Source of Photograph:

Own photograph, captured via GPS Map Camera (GPS-enabled).

Place:

Jimmy Carter Road, VIT University, Vellore, Tamil Nadu 632014, India (Lat 12.971415, Long 79.164106)

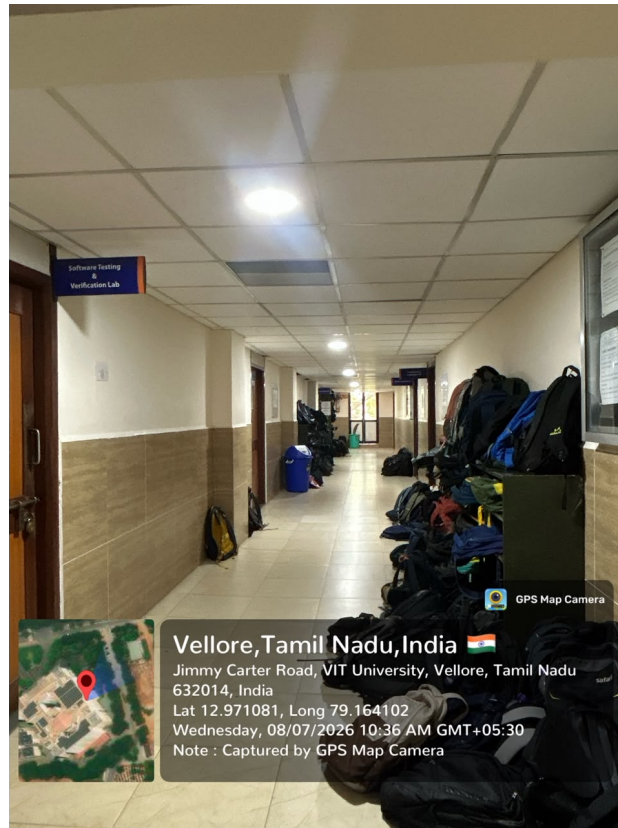
Reasons why the design is bad:

- No visible emergency signage indicating the staircase's purpose
- No visible lighting for nighttime/emergency use
- Narrow structure could bottleneck during mass evacuation
- Overhanging tree branches obstruct part of the structure

Suggested Improvement:

- Add illuminated "Fire Exit" signage at each level
- Install emergency lighting along the staircase
- Trim surrounding vegetation for clear access
- Widen or add a second escape route for large occupancy buildings

Bad Design 5: No Dedicated Storage for Bags Outside Lab



Description:

A corridor outside a lab (Software Testing & Verification Lab) where dozens of backpacks are piled on the floor and stacked on a small improvised stand, since no lockers or storage racks are provided.

Source of Photograph:

Own photograph, captured via GPS Map Camera (GPS-enabled).

Place:

Jimmy Carter Road, VIT University, Vellore, Tamil Nadu 632014, India (Lat 12.971081, Long 79.164102)

Reasons why the design is bad:

- Bags obstruct the walking path, creating a trip/fire hazard
- No designated storage forces students to improvise, creating clutter
- Reduces corridor width available for movement/evacuation
- Unattended bags in open corridor raise security/theft concerns

Suggested Improvement:

- Install proper lockers or a bag rack outside the lab
- Mark a designated bag-storage zone away from the walking path
- Add signage directing students to use provided storage

Bad Design 6: Exposed Outdoor Switch/Socket Board with Loose Wiring



Description:

An outdoor switch and socket board mounted directly on a pole at chest height, holding two switches, a multi-pin socket, and a power plug. The wiring dangles loosely from the bottom of the board, unshielded from rain and dust, and the unit sits right beside an active construction barricade.

Source of Photograph:

Own photograph, captured via GPS Map Camera (GPS-enabled).

Place:

Jimmy Carter Road, VIT University, Vellore, Tamil Nadu 632014, India (Lat 12.971112, Long 79.164356)

Reasons why the design is bad:

- Exposed wiring is a safety hazard, especially in rain
- No protective housing or weatherproof cover on the board
- Positioned in a high-traffic zone next to construction barricades
- Sockets exposed to dust/moisture, reducing safety and lifespan

Suggested Improvement:

- Install a weatherproof, IP65-rated enclosure
- Route all wiring through conduit instead of leaving it loose
- Relocate the board away from the active construction/pedestrian zone
- Add a lockable cover or door to the board

Bad Design 7: Overgrown Enclosure Blocking Fire Hose Cabinet and Water Station



Description:

A steel outdoor drinking-water/wash station with multiple taps sits inside a small fenced enclosure that has become overgrown with trees and shrubs. A fire hose cabinet is mounted directly behind the station, with vegetation partially blocking access to it. Personal items (a bottle, rags) are left cluttered on top of the station.

Source of Photograph:

Own photograph, captured via GPS Map Camera (GPS-enabled).

Place:

Jimmy Carter Road, VIT University, Vellore, Tamil Nadu 632014, India (Lat 12.971285, Long 79.163509)

Reasons why the design is bad:

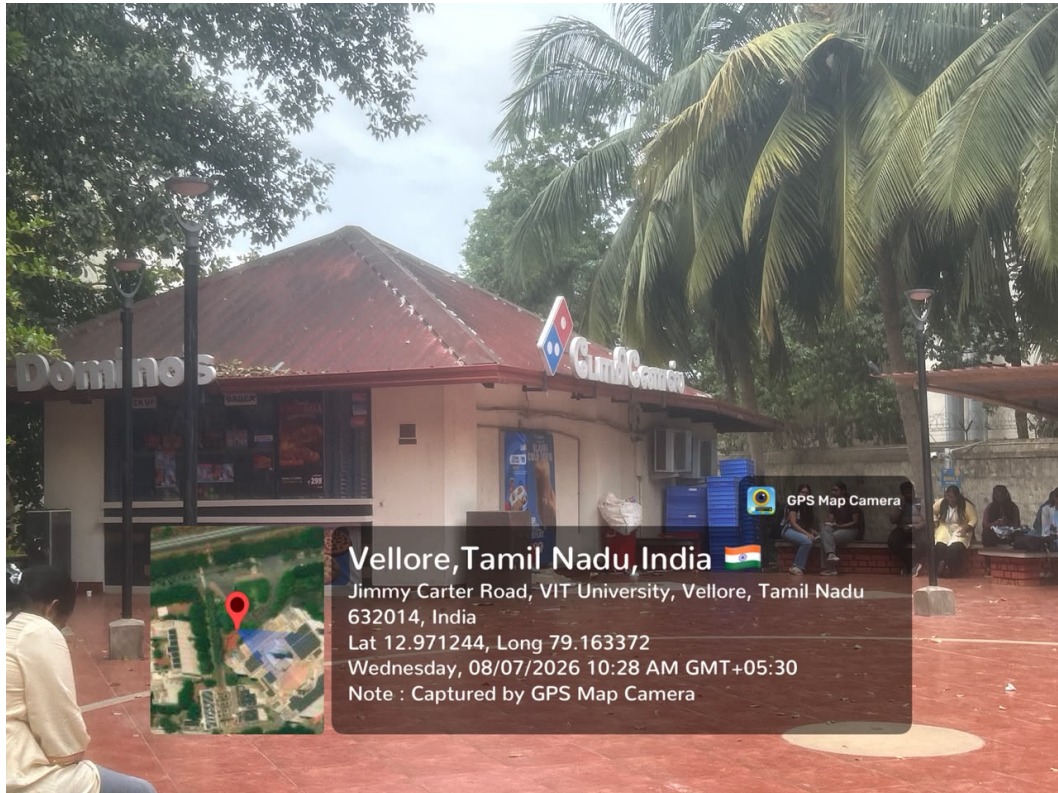
- Overgrown vegetation obstructs quick access to the fire hose cabinet in an emergency
- Water taps positioned close to landscaping risk debris/leaves entering the water
- Clutter left on the station is unhygienic for a shared drinking-water point
- Unmaintained fenced enclosure can trap stagnant water and encourage mosquitoes

Suggested Improvement:

- Regularly trim vegetation to keep fire safety equipment clearly accessible
- Maintain a clear boundary between the drinking-water station and landscaping
- Enforce cleanliness and disallow personal items on the shared station

- Maintain fencing and drainage within the enclosure

Bad Design 8: Cluttered Storage and AC Unit Blocking Kiosk Walkway



Description:

A food kiosk (Domino's) in an outdoor food court has a stack of blue storage crates and an external AC condenser unit placed directly beside the customer-facing walkway and seating area, narrowing the space available for pedestrians.

Source of Photograph:

Own photograph, captured via GPS Map Camera (GPS-enabled).

Place:

Jimmy Carter Road, VIT University, Vellore, Tamil Nadu 632014, India (Lat 12.971244, Long 79.163372)

Reasons why the design is bad:

- Stacked crates and the AC unit occupy the customer walkway, reducing usable space
- Creates a tripping hazard for pedestrians moving around the crates
- AC condenser heat/noise affects the comfort of people seated or walking nearby
- Unorganized outdoor storage gives an untidy, unprofessional appearance

Suggested Improvement:

- Relocate crates to a dedicated storage area behind or beside the kiosk
- Position the AC unit away from the pedestrian path, or screen it off
- Mark a clear, unobstructed walkway boundary
- Maintain regular housekeeping of the outdoor storage area